

# Supplement S3: Observable Dataset

**Complete Observable Dataset (95 entries) + Per-Sector Relation Tables** Brieuc de La Fournière

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## S3.1 Introduction

This supplement provides the complete observable dataset of the  $K_7$  framework: 95 observables derived from a single  $G_2$  metric (169 Chebyshev-Cholesky geometric parameters, zero freely adjustable physical parameters), organized by derivation type and cross-referenced with Lean formal verification. The dataset includes 2 BH remnant predictions from Pinčák et al. 2026 (main text ref. [42], classified Type IV) and 6 Type IV metric block eigenvalues (v3.4.29 ledger).

The 95 observables decompose as **33(I) + 19(II) + 21(III) + 22(IV)** across four types of increasing derivation complexity. All predictions trace to a single compact  $G_2$  manifold  $K_7$  with  $(b_2, b_3) = (21, 77)$  and  $E_8 \times E_8$  gauge structure, 169 metric parameters, zero continuously adjustable physical parameters. The pair  $(b_2, b_3) = (21, 77)$  does not appear among the catalogued compact  $G_2$  manifolds; the Joyce-Karigiannis  $Z_2^3$  route realizes it at the topological/lattice level (S1 §8.4), the companion scheme [E] discharges the datum-level analytic layer conditionally, and the global-analytic completion remains open (two-layer boundary: main §9). The certified metric provides computational evidence for a new  $G_2$  manifold.

The dataset serves as the canonical reference for all framework predictions, superseding results previously scattered across internal working supplements. Every entry traces back to a specific computation, a JSON artifact, and (where applicable) a Lean theorem.

## S3.2 Type Classification

Observables are classified into four types by derivation directness. The type assignment reflects the number of physical identification steps between the  $G_2$  metric and the final prediction:

| Type | Count | Description                                | Derivation chain                                          | Lean coverage |
|------|-------|--------------------------------------------|-----------------------------------------------------------|---------------|
| I    | 33    | Direct algebraic from structural constants | topology $\rightarrow$ formula $\rightarrow$ ratio        | 33/33 (100%)  |
| II   | 19    | One physical identification step           | formula + VEV or scale $\rightarrow$ dimensional quantity | 0/19 (0%)     |

| Type | Count | Description                    | Derivation chain                                        | Lean coverage |
|------|-------|--------------------------------|---------------------------------------------------------|---------------|
| III  | 21    | Multi-step dynamical chains    | metric $\rightarrow$ mechanism $\rightarrow$ observable | 14/21 (67%)   |
| IV   | 22    | Structural/internal quantities | metric $\rightarrow$ diagnostic                         | 8/22 (36%)    |

**Type I** (33 observables): Direct from metric geometry and topological invariants ( $b_2$ ,  $b_3$ ,  $\dim(G_2)$ ,  $\dim(E_8)$ , etc.). These are dimensionless ratios expressed as algebraic combinations of integers. Examples:  $\sin^2\theta_W = 3/13$ ,  $Q_{\text{Koide}} = 2/3$ ,  $m_\tau/m_e = 3477$ . All 33 are Lean-certified. The two BH remnant topological predictions are classified as Type IV structural diagnostics (not part of the 33 Type I Lean-certified core).

**Type II** (19 observables): One physical identification step beyond Type I. Absolute masses from ratios  $\times$  VEV (e.g.,  $m_u = (m_u/m_d) \times m_{d,\text{exp}}$ ), CKM magnitudes from Wolfenstein parametrization, extended quark ratios. From the `gift_observables.csv` pipeline.

**Type III** (21 observables): Multi-step chains involving dynamical mechanisms: - wilson\_line non-adiabatic eigenvalue splitting (3 observables: raw lepton ratios) - S11 RGE gauge coupling running (4 observables:  $\sin^2\theta_W$ ,  $\alpha_{\text{em}}^{-1}$ ,  $\alpha_s$  at  $M_Z$ ,  $M_{\text{GUT}}$ ) - spectral 7D spectral geometry (5 observables: Weyl exponent, KK states, fiber channels) - gauge\_bundle gauge bundle data (4 observables:  $f_{\text{IJ}}$  conditioning,  $\alpha_{\text{ratio}}$ , Yukawa rank) - instanton instanton volumes + combined wilson\_line+instanton (5 observables:  $\Delta V$ 's, combined ratios)

**Type IV** (22 observables): Structural and internal consistency quantities with no direct experimental comparison. These include topology diagnostics ( $b_2$ ,  $b_3$ ,  $\chi(K_7)$ ), Newton-Kantorovich certification values ( $h$ , distance, margin), Gram matrix conditioning ( $\text{cond}(G_{K3})$ ,  $\text{cond}(G_{77})$ ), spectral counts, metric block eigenvalues ( $g_{ss} = 19/6$ ,  $g_{\{T^2\}} = 7/6$ ,  $g_{K3} \approx 64/77$ ), and Pinčák et al. 2026 (main text ref. [42]) topological diagnostics ( $N_{\text{QNM}} = 98$ ,  $M_{\text{res}}$ , instanton suppression  $\times 77$ ).  $M_{\text{res}}$  and  $N_{\text{QNM}}$  are the two BH-remnant-related entries; they are classified here as structural diagnostics, not as Type I or Type III, because they require a physical scale identification ( $\tau_0 = v_{EW}$ ) not derivable from topology alone. These quantities verify internal consistency rather than predict measurements directly.

**Derivation depth:** The type assignment correlates with derivation chain length. Type I observables require 1 algebraic step; Type II requires 2 steps (algebra + scale identification); Type III requires 3–5 steps (algebra + dynamics + calibration); Type IV involves diagnostic computation chains of variable length.

## S3.3 Methodology

### Experimental Sources

All experimental reference values are drawn from four standard compilations:

| Source                                  | Coverage                                                     | Date     |
|-----------------------------------------|--------------------------------------------------------------|----------|
| <b>PDG 2024</b> [1]                     | Particle masses, gauge couplings, CKM elements               | 2024     |
| <b>NuFIT 6.0</b> [27]                   | Neutrino oscillation parameters (NO w/o SK)                  | Oct 2024 |
| <b>Planck PR4</b> (Tristram+ 2024) [49] | Cosmological parameters: $h$ , $\sigma_8$ , $\Omega_\Lambda$ | 2024     |
| <b>CODATA 2022</b> [50]                 | Lepton mass ratios                                           | 2022     |

Selected NuFIT 6.0 values:  $\sin^2\theta_{23} = 0.561$ ,  $\delta_{\text{CP}} = 177^\circ$ . Selected Planck PR4 values:  $h = 0.6764$ ,  $\sigma_8 = 0.807$ ,  $\Omega_\Lambda = 0.6847$ . NuFIT 6.0 is the *frozen dataset* for all deviation statistics in this supplement; the subsequent NuFIT 6.1 release (2025,  $\delta_{\text{CP}} = 207^\circ +23/-20$  NO w/o SK-atm) is tracked in the main paper §4.2 and moves the  $\delta_{\text{CP}}$  prediction inside the  $1\sigma$  band.

## Structural Constants

The 20 structural constants from which all 95 observables derive:

| # | Constant           | Value | Origin                                                                                                                                                      |
|---|--------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | $\dim(E_8)$        | 248   | $E_8$ Lie algebra                                                                                                                                           |
| 2 | $\text{rank}(E_8)$ | 8     | Cartan subalgebra                                                                                                                                           |
| 3 | $\dim(G_2)$        | 14    | $\text{Aut}(\mathbb{O})$                                                                                                                                    |
| 4 | $\dim(K_7)$        | 7     | $\text{Im}(\mathbb{O})$                                                                                                                                     |
| 5 | $b_2(K_7)$         | 21    | Input; confirmed by spectral analysis (Paper B [B], §2.3). Any Mayer-Vietoris decomposition is conditional on building block identification (open problem). |
| 6 | $b_3(K_7)$         | 77    | Input; confirmed by spectral analysis (Paper B [B], §2.3). Any Mayer-Vietoris decomposition is conditional on building block identification (open problem). |
| 7 | $N_{\text{gen}}$   | 3     | Index theorem                                                                                                                                               |
| 8 | $p_2$              | 2     | $\dim(G_2)/\dim(K_7)$                                                                                                                                       |

| #  | Constant                | Value | Origin                                   |
|----|-------------------------|-------|------------------------------------------|
| 9  | Weyl                    | 5     | Triple identity                          |
| 10 | $H^*$                   | 99    | $b_2 + b_3 + 1$                          |
| 11 | $\dim(J_3(\mathbb{O}))$ | 27    | Jordan algebra                           |
| 12 | $\dim(E_6)$             | 78    | $E_6$ Lie algebra                        |
| 13 | $\dim(E_7)$             | 133   | $E_7$ Lie algebra                        |
| 14 | $\dim(F_4)$             | 52    | $\text{Aut}(J_3(\mathbb{O}))$            |
| 15 | $\text{fund}(E_7)$      | 56    | $E_7$ fundamental                        |
| 16 | $ \text{PSL}(2,7) $     | 168   | Fano automorphisms                       |
| 17 | $D_{\text{bulk}}$       | 11    | $4+7$                                    |
| 18 | $\alpha_{\text{sum}}$   | 13    | $\text{rank}(E_8) + \text{Weyl}$         |
| 19 | $\det(g)_{\text{den}}$  | 32    | $b_2 + \dim(G_2) - N_{\text{gen}}$       |
| 20 | $\det(g)_{\text{num}}$  | 65    | $\text{Weyl} \times \alpha_{\text{sum}}$ |

## Consolidation Pipeline

The master dataset (`gift_observables.json`) is generated by `observable_dataset.py`, which collects the individual JSON results of the internal computation supplements and performs 5 verification checks:

1. **Completeness:** All 95 entries present across 4 types
2. **Uniqueness:** No duplicate observable IDs
3. **Consistency:** Predicted values match source JSONs to machine precision
4. **Source tracing:** Every entry maps to a documented computation step
5. **Format validation:** JSON, CSV, and LaTeX outputs agree

Each observable record contains: prediction (float64), fraction (exact symbolic), experimental value and uncertainty, experimental source, relative deviation (%), Lean theorem name (if certified), and provenance (computation step).

## S3.4 Complete Observable Table

**Table S3.1:** Complete observable dataset (95 entries, v3.4.29 ledger). Columns: name, prediction, experimental value, relative deviation (%), type (I/II/III/IV), source computation, Lean verification status.

[The full LaTeX table is provided in `data/gift_observables_latex.tex` and will be included at typesetting.]

## Per-Sector Relation Tables

*The per-sector Type I tables and their derivational prose moved here from main §4 at 3.5; the main paper keeps a representative summary (main §4.1) and the  $\delta_{CP}$  discussion (main §4.2). Conventions: PDG*

2024 [1]; NuFIT 6.0 frozen dataset [27]; Planck PR4 [49].

### Gauge sector

$$\sin^2 \theta_W = \frac{b_2}{b_3 + \dim(G_2)} = \frac{21}{91} = \frac{3}{13} = 0.230769$$

Experimental (PDG 2024) [1]:  $0.23122 \pm 0.00004$ . Deviation: **0.195%**. The numerator  $b_2$  counts gauge moduli; the denominator  $b_3 + \dim(G_2)$  counts matter plus holonomy degrees of freedom. The ratio measures gauge-matter coupling geometrically.

$$\alpha_s(M_Z) = \frac{\sqrt{2}}{\dim(G_2) - p_2} = \frac{\sqrt{2}}{12} = 0.11785$$

Experimental:  $0.1180 \pm 0.0009$ . Deviation: **0.126%**.

### Lepton sector

**Koide parameter.** The Koide relation among charged lepton masses,  $Q = (m_e + m_\mu + m_\tau)/(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2 = 2/3$ , has resisted explanation since 1982. Using contemporary mass values it holds to six significant figures:  $Q_{\text{exp}} = 0.666661 \pm 0.000007$ . The  $K_7$  framework provides  $Q_{\text{Koide}} = \dim(G_2)/b_2 = 14/21 = 2/3$ , an algebraic identity in two topological invariants; no fitting is involved.

| Approach                          | Result                                              | Status                    |
|-----------------------------------|-----------------------------------------------------|---------------------------|
| Preon models (Koide 1982)         | $Q = 2/3$ assumed                                   | Circular                  |
| $S_3$ symmetry (various)          | $Q \sim 2/3$ fitted                                 | Approximate               |
| <b><math>K_7</math> framework</b> | <b><math>Q = \dim(G_2)/b_2 = 14/21 = 2/3</math></b> | <b>Algebraic identity</b> |

Deviation: **0.0009%**, the most precise agreement in the framework.

**Tau-electron mass ratio.**  $m_\tau/m_e = \dim(K_7) + 10 \times \dim(E_8) + 10 \times H^* = 7 + 2480 + 990 = 3477$ . Experimental:  $3477.15 \pm 0.05$ ; deviation **0.004%**. The integer  $3477 = 3 \times 19 \times 61 = N_{\text{gen}} \times \text{prime}(8) \times \kappa_T^{-1}$  factorizes into framework constants.

**Muon-electron mass ratio.**  $m_\mu/m_e = \dim(J_3(\mathbb{O}))^\wedge \varphi = 27^\wedge \varphi = 207.01$ , with  $\varphi = (1 + \sqrt{5})/2$  from the McKay correspondence  $E_8 \leftrightarrow 2I$  (binary icosahedral group). Experimental: 206.768; deviation **0.118%**. See the golden-ratio caveat in main §4.1:  $\varphi$  is the only non-integer input among the 33 Type I relations, and its status is weaker than the integer-ratio derivations (S2 §11).

### Quark sector

$$\frac{m_s}{m_d} = p_2^2 \times \text{Weyl} = 4 \times 5 = 20$$

Experimental (PDG 2024):  $20.0 \pm 1.0$ . Deviation: **0.00%**.

$$\frac{m_b}{m_t} = \frac{b_0}{2b_2} = \frac{1}{42}$$

Experimental:  $0.024 \pm 0.001$ . Deviation: **0.79%**. The constant  $42 = p_2 \times N_{\text{gen}} \times \dim(K_7) = 2 \times 3 \times 7$  is a structural invariant (not to be confused with  $\chi(K_7) = 0$ , which vanishes for any compact odd-dimensional manifold).

### Neutrino mixing angles

| Angle         | Formula                                         | Prediction    | NuFIT 6.0 [27]         | Dev.  |
|---------------|-------------------------------------------------|---------------|------------------------|-------|
| $\theta_{12}$ | $\arctan(\sqrt{(\delta/\gamma_{\text{GIFT}})})$ | $33.40^\circ$ | $33.41 \pm 0.75^\circ$ | 0.03% |
| $\theta_{13}$ | $\pi/b_2$                                       | $8.57^\circ$  | $8.54 \pm 0.12^\circ$  | 0.37% |
| $\theta_{23}$ | $\arcsin((b_3 - p_2)/H^*)$                      | $49.25^\circ$ | $49.3 \pm 1.0^\circ$   | 0.10% |

Auxiliary parameters:  $\delta = 2\pi/\text{Weyl}^2 = 2\pi/25$  and  $\gamma_{\text{GIFT}}$  (historical symbol name)  $= (2 \times \text{rank}(E_8) + 5 \times H^*)/(10 \times \dim(G_2) + 3 \times \dim(E_8)) = 511/884$ . The  $\delta_{\text{CP}}$  prediction is discussed in main §4.2.

### Higgs sector

$$\lambda_H = \frac{\sqrt{\dim(G_2) + N_{\text{gen}}}}{\det(g)_{\text{den}}} = \frac{\sqrt{17}}{32} = 0.1289$$

The Higgs self-coupling combines holonomy dimension with generation count, normalized by the metric determinant scale. Experimental (PDG 2024):  $0.129 \pm 0.001$ . Deviation: **0.12%**. A TCS alternative  $\lambda_{\text{H}} = b_2(M_1)/(b_3 + b_2(M_2)) = 11/87 = 0.1264$  is purely rational (see S2 §17).

### Boson mass ratios

| Observable                  | Formula                                          | Prediction | Experimental        | Dev.  |
|-----------------------------|--------------------------------------------------|------------|---------------------|-------|
| $m_{\text{H}}/m_{\text{W}}$ | $(N_{\text{gen}} + \dim(E_6))/\dim(F_4) = 81/52$ | 1.5577     | $1.558 \pm 0.002$   | 0.02% |
| $m_{\text{W}}/m_{\text{Z}}$ | $(2b_2 - \text{Weyl})/(2b_2) = 37/42$            | 0.8810     | $0.8815 \pm 0.0002$ | 0.06% |
| $m_{\text{H}}/m_{\text{t}}$ | $\text{fund}(E_7)/b_3 = 56/77$                   | 0.7273     | $0.725 \pm 0.003$   | 0.31% |

### CKM matrix

| Observable                        | Formula                                            | Prediction | Experimental        | Dev.  |
|-----------------------------------|----------------------------------------------------|------------|---------------------|-------|
| $\sin^2(\theta_{12} \text{ CKM})$ | $\text{fund}(E_7)/\text{dim}(E_8)$<br>$= 56/248$   | 0.2258     | $0.2250 \pm 0.0006$ | 0.36% |
| $A_{\text{Wolfenstein}}$          | $(\text{Weyl} + \text{dim}(E_6))/H^* =$<br>$83/99$ | 0.838      | $0.836 \pm 0.015$   | 0.29% |
| $\sin^2(\theta_{23} \text{ CKM})$ | $\text{dim}(K_7)/\text{PSL}(2,7)$<br>$= 7/168$     | 0.0417     | $0.0412 \pm 0.0008$ | 1.13% |

The Cabibbo angle emerges from the ratio of the  $E_7$  fundamental representation to  $E_8$  dimension.

### Cosmological observables

| Observable                             | Formula                                                     | Prediction | Experimental        | Dev.   |
|----------------------------------------|-------------------------------------------------------------|------------|---------------------|--------|
| $\Omega_{\text{DM}}/\Omega_{\text{b}}$ | $(1 + 2b_2)/\text{rank}(E_8) =$<br>$43/8$                   | 5.375      | $5.375 \pm 0.1$     | 0.00%  |
| $n_s$                                  | $\zeta(11)/\zeta(5)$                                        | 0.9649     | $0.9649 \pm 0.0042$ | 0.004% |
| $h$ (Hubble)                           | $(\text{PSL}(2,7) - 1)/\text{dim}(E_8) =$<br>$167/248$      | 0.6734     | $0.674 \pm 0.005$   | 0.09%  |
| $\Omega_{\text{b}}/\Omega_{\text{m}}$  | $\text{Weyl}/\det(g)_{\text{den}} =$<br>$5/32$              | 0.1562     | $0.157 \pm 0.003$   | 0.16%  |
| $\sigma_8$                             | $(p_2 + 32)/(2b_2) =$<br>$34/42$                            | 0.8095     | $0.811 \pm 0.006$   | 0.18%  |
| $\Omega_{\text{DE}}$                   | $\ln(2) \times (b_2 + b_3)/H^* =$<br>$\ln(2) \times 98/99$  | 0.6861     | $0.6847 \pm 0.005$  | 0.21%  |
| $Y_{\text{p}}$                         | $(1 + \text{dim}(G_2))/\kappa_{\text{T}}^{-1} =$<br>$15/61$ | 0.2459     | $0.245 \pm 0.003$   | 0.37%  |

The dark-to-baryonic matter ratio  $\Omega_{\text{DM}}/\Omega_{\text{b}} = 43/8$  is exact. The structural invariant  $2b_2 = 42$  that gives  $m_{\text{b}}/m_{\text{t}} = 1/42$  also determines this cosmological ratio, connecting quark physics to large-scale structure through  $K_7$  geometry.

### Sector-by-Sector Performance

The 66 observables with experimental comparison (Types I+II+III) organize into sectors of varying precision:

**Best-performing sectors** (mean dev < 0.5%): - **Lepton** (3 obs): mean 0.04%:  $Q_{\text{Koide}}$  (0.001%),  $m_{\tau}/m_e$  (0.004%),  $m_{\mu}/m_e$  (0.12%) - **Electroweak** (3 obs): mean 0.11%,  $\alpha^{-1}$  (0.002%),  $\alpha_s$  (0.126%),

$\sin^2\theta_W$  (0.20%) - **Boson** (3 obs): mean 0.13%,  $m_H/m_W$  (0.02%),  $m_W/m_Z$  (0.06%),  $m_H/m_t$  (0.31%) - **Cosmology** (7 obs): mean 0.15%,  $n_s$  (0.004%),  $h$  (0.09%),  $\Omega_b/\Omega_m$  (0.16%),  $\sigma_8$  (0.18%),  $\Omega_{DE}$  (0.21%),  $Y_p$  (0.37%),  $\Omega_{DM}/\Omega_b$  (0.00%) - **Quark** (4 obs): mean 0.24%,  $m_s/m_d$  (0.00%),  $m_u/m_d$  (0.05%),  $m_c/m_s$  (0.12%),  $m_b/m_t$  (0.79%) - **PMNS** (4 obs): mean 0.29%,  $\theta_{12}$  (0.03%),  $\theta_{23}$  (0.10%),  $\sin^2\theta_{12}$  (0.23%),  $\sin^2\theta_{13}$  (0.81%)

**Good sectors** (mean dev 0.5–3%): - **CKM** (3 obs): mean 0.59%:  $A_{Wolf}$  (0.29%),  $\sin^2\theta_{12}$  (0.36%),  $\sin^2\theta_{23}$  (1.13%) - **Gauge running** (4 obs): mean 2.3%,  $\sin^2\theta_W(RGE)$  (2.78%),  $\alpha_{em}^{-1}(RGE)$  (2.53%),  $\alpha_s(RGE)$  (3.7%),  $M_{GUT}$  (0.00%)

**Challenging sectors** (mean dev  $> 3\%$ ): - **Instanton** (2 obs): mean 10.9%:  $\Delta V(e-\tau)$  (5.9%),  $\Delta V(e-\mu)$  (15.9%); both reduced to sub-percent by the combined pipeline (§S3.5) - **Bundle** (1 obs):  $\kappa(gauge)$  (4.7%), classified with the gauge-bundle diagnostics as in main §5.5

**B-test consistency** (derived, not counted as independent observable): The MSSM B parameter  $B = (\alpha_1^{-1} - \alpha_2^{-1})/(\alpha_2^{-1} - \alpha_3^{-1})$  evaluates to 1.4033 using the framework couplings (0.23% from the theoretical 7/5), closer than the purely experimental  $B = 1.3948$  (0.37%). The identity  $B = 7/5$  is equivalent to  $\alpha_{em}^{-1}(M_Z) = 91\sqrt{2} = \dim(\Lambda^2 \mathfrak{g}_2) \cdot \sqrt{2}$  and implies the holonomy sequence  $\alpha_1^{-1}:\alpha_2^{-1}:\alpha_3^{-1} = \dim(G_2):\dim(K_7):p_2 = 14:7:2$ . See §5.4 of the main text.

## Exact Matches

11 observables achieve deviations below 0.01% (effectively exact):

| Observable          | Prediction         | Exp.               | Dev.   |
|---------------------|--------------------|--------------------|--------|
| $\alpha^{-1}$       | 137.033            | 137.036            | 0.002% |
| $Q_{Koide}$         | 2/3                | 0.666661           | 0.001% |
| $m_\tau/m_e$        | 3477               | 3477.15            | 0.004% |
| $m_s/m_d$           | 20                 | 20.0               | 0.000% |
| $n_s$               | 0.9649             | 0.9649             | 0.004% |
| $m_u$               | 2.16               | 2.16               | 0.00%  |
| $ V_{tb} $          | 0.999              | 0.999              | 0.00%  |
| $m_c$               | 1.27               | 1.27               | 0.00%  |
| $M_{GUT}$           | $2 \times 10^{16}$ | $2 \times 10^{16}$ | 0.00%  |
| $\alpha_{GUT}^{-1}$ | 25.3               | 25.3               | 0.00%  |
| $\text{rank}(Y)$    | 3                  | 3                  | 0.00%  |

Note:  $\delta_{CP}$  canonical prediction is  $197^\circ$  (11.3% from the frozen NuFIT 6.0 central; 4.8% from the NuFIT 6.1 central, inside  $1\sigma$ ); a compactification factor 62/69 brings it to  $177.014^\circ$  vs the 6.0 central (0.008%) but is not adopted, see main §4.2.  $m_c/m_s$  deviation is 0.12% and its Type I product with  $m_s/m_d$  gives  $m_c/m_d = 234.3$  vs the PDG scale-common ratio 234 (0.12%), so neither is listed among the exact matches above; the mixed-scale value 272 that would arise from combining  $m_c(m_c)$  and  $m_d(2 \text{ GeV})$  violates the §4 convention (all quark ratios at the PDG reference scales) and is not tracked.



## Known Outliers

Two observables deviate by more than 5%, each with an identified physical origin:

**$\delta\_CP$ :** The framework prediction is  $197^\circ$  (pure topological:  $7 \times 14 + 99$ ), deviating 11.3% from the frozen NuFIT 6.0 central ( $177^\circ \pm 20^\circ$ , edge of  $1\sigma$ ) and 4.8% from the NuFIT 6.1 central ( $207^\circ +23/-20$ , inside  $1\sigma$ ). PSLQ analysis identifies a compactification factor 62/69 (§4.2 of main paper), documented as a structural observation but not adopted: the canonical prediction remains  $197^\circ$ . The experimental central value drifted from  $197^\circ$  (NuFIT 5.2) to  $177^\circ$  (NuFIT 6.0) to  $207^\circ$  (NuFIT 6.1), bracketing the prediction. Falsifiable by DUNE (first beam targeted 2031; physics run late 2030s–2040s).

**$\alpha\_s(RGE)$  = +3.7%:**  $G_2$ -MSSM split-spectrum matching (MSSM above  $M\_squark = 3165$  GeV, SM + gauginos below ( $b_3 = -5$ )) gives  $\alpha\_s = 0.1224$  (exp 0.1180, dev 3.7%). A naive degenerate-spectrum treatment would give 12.1%. The topological value  $\alpha\_s = \sqrt{2}/12 = 0.11785$  (Type I) matches experiment to 0.126%.

**$\Delta V(e-\mu)$  = +15.9%:** The instanton volume assignment  $\Delta V(e-\mu) = 3.271$  versus target  $\ln(16.82) = 2.823$  reflects the optimal assignment problem for 57 associative cycles. The combined wilson\_line+instanton pipeline with  $\alpha = e^K$  (§S3.5) reduces this to 0.75%.

**$\kappa(gauge)$  = +4.7%:** The gauge kinetic function conditioning  $\text{cond}(f\_IJ) = 1.047$  measures departure from exact universality. The deviation from 1.000 reflects  $K_7$  geometry at the percent level.

## Summary Statistics

| Metric                       | Value                             |
|------------------------------|-----------------------------------|
| Total observables            | 95                                |
| Type breakdown               | 33(I) + 19(II) + 21(III) + 22(IV) |
| With experimental comparison | 66                                |
| Exact matches (dev < 0.01%)  | 11                                |
| Within 1% of experiment      | 53                                |
| Within 5% of experiment      | 63                                |
| Lean-certified               | 55                                |
| Structural constants         | 20                                |
| Metric parameters            | 169                               |

## S3.5 The Combined wilson\_line+instanton Pipeline

The lepton mass hierarchy (combined lepton ratio observables) uses two complementary geometric mechanisms. The complete derivation is in §6 of the main text; results are summarised here for cross-reference.

Raw wilson\_line ( $K_3$  fiber eigenvalue splitting at  $c = 0.452$ ):  $m_\tau/m_e = 3403$  (2.1%),  $m_\tau/m_\mu = 16.54$  (1.7%),  $m_\mu/m_e = 205.7$  (0.5%). Raw instanton (associative 3-cycle volumes, 57 cycles):  $\Delta V(e-\tau)$

$= 8.633$  (5.9%),  $\Delta V(e-\mu) = 3.271$  (15.9%). Combined with  $\alpha = e^{\{K_0\}} = V^{\{-3\}} = 0.002763$  (Kähler potential of  $K_7$ , zero free parameters):

- $m_\tau/m_e = 3485$  (exp 3477, **0.23%**)
- $m_\tau/m_\mu = 16.69$  (exp 16.82, **0.75%**)
- $m_\mu/m_e = 208.8$  (exp 206.8, **0.98%**)

Certified in Lean: `AssociativeVolumes.lean` (14 conjuncts, 0 sorry). Full mechanism description: main §6.4.

### S3.6 Sensitivity Cross-Reference

The complete sensitivity analysis is in §7 of the main text. Key results for cross-reference:

- **Effective DOF:**  $r_{\text{eff}} = 15.53$  (SVD of  $20 \times 33$  constant-usage Jacobian)
- **Overdetermination:**  $33/15.53 = 2.13\times$  (more constraints than free dimensions)
- **Cross-coupling:** 155/528 observable pairs share a structural constant; all form one connected component
- **P(coincidence, uniform)**  $= 10^{-346}$  ( $\chi^2$  + Fisher combined; archived diagnostic, see Supplement S4)
- **P(coincidence, algebraic)**  $= 10^{-133}$  (4.2M random formulas from same 20 constants; archived diagnostic, see Supplement S4). These joint-probability figures are internal-consistency diagnostics, superseded as headline methodology by the Sieve reading (main §7).

Figures: `fig_sensitivity_heatmap.png`, `fig_constant_usage.png`, `fig_observable_correlations.png`, `fig_mc_per_observable.png`.

### S3.7 Lean Cross-References

Of the 95 observables, **55 are formally verified** in Lean 4 across the following certificate files:

| Lean File                     | Axioms | Theorems | Conjuncts | Coverage                               |
|-------------------------------|--------|----------|-----------|----------------------------------------|
| <code>Foundations.lean</code> | 0*     | n/a      | 39        | Type I: metric, torsion, topology      |
| <code>Predictions.lean</code> | 0*     | n/a      | 56        | Types I–III: couplings, masses, mixing |

| Lean File                                     | Axioms | Theorems | Conjuncts | Coverage                                                                                  |
|-----------------------------------------------|--------|----------|-----------|-------------------------------------------------------------------------------------------|
| <code>Spectral.lean</code>                    | 0*     | n/a      | 45        | Types III-IV: KK spectrum, Weyl law                                                       |
| <code>Metric Eigenvalues.lean</code>          | 0      | n/a      | 15        | Metric fractions: $g_{ss}=19/6$ , $g_{T^2}=7/6$ , $\gamma^2=24\pi^2/7$ ( $T^2$ Laplacian) |
| <code>Spectral Invariants.lean</code>         | 0      | n/a      | 10        | Heat kernel, $\zeta'(0)$ , $b_1=0$                                                        |
| <code>Compactification Correction.lean</code> | 0      | n/a      | 6         | $\delta_{CP}$ compactification factor                                                     |
| <code>TCSGaugeBreaking.lean</code>            | 0      | 14       | 10        | Type III: gauge breaking chain                                                            |
| <code>GaugeBundleData.lean</code>             | 0      | 12       | 11        | Type III: bundle universality                                                             |
| <code>Associative Volumes.lean</code>         | 0      | 19       | 14        | Type III: instanton hierarchy                                                             |
| <code>ComputedWeylLaw.lean</code>             | 0      | 8        | 7         | Type III: 7D spectral geometry                                                            |

\*4 external data-package axioms (of 15 classified in the A–F taxonomy of the v3.4.29 release audit; see main §8.1 for the reconciliation). All substantive: standard theorems (Cheeger inequality) + geometric structure (TCS spectral bounds) + physical inputs (literature package: CGN+Joyce).  $G_2$  group structure proven by `native_decide`.

**Total certificate:** 213 conjuncts (top-level conjunctions of the master certificates:  $39 + 56 + 45 + 15 + 10 + 6 + 10 + 11 + 14 + 7$ ), 0 sorry, 146 .lean files under **GIFT/** (140 core + 6 generated), 8394 build jobs (Lean 4.29.0).

### Coverage by Type

| Type         | Certified | Total     | Coverage   |
|--------------|-----------|-----------|------------|
| <b>I</b>     | 33        | 33        | 100%       |
| <b>II</b>    | 0         | 19        | 0%         |
| <b>III</b>   | 14        | 21        | 67%        |
| <b>IV</b>    | 8         | 22        | 36%        |
| <b>Total</b> | <b>55</b> | <b>95</b> | <b>58%</b> |

**Type II uncertified:** The 19 Type II observables derive from the CSV pipeline (absolute masses from

ratio  $\times$  VEV, CKM magnitudes from Wolfenstein). These involve dimensional quantities and intermediate data that have not yet been axiomatized in Lean. The underlying Type I ratios from which they derive are all certified.

**Type IV partial:** 8 of 22 structural quantities are certified (Betti numbers, holonomy dimension, NK certification bounds). The remaining 14 involve spectral counts, conditioning numbers, metric eigenvalues, and BH remnant estimates that require floating-point Lean infrastructure not yet developed.

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### S3.8 Data Availability

The complete dataset is generated deterministically by the `observable_dataset.py + deviation_systematics.py` pipeline from the frozen inputs and is available in three formats:

- **JSON:** `gift_observables.json` (machine-readable, full metadata per observable)
- **CSV:** `gift_observables.csv` (tabular, 95 rows  $\times$  13 columns, for analysis)
- **LaTeX:** `gift_observables_latex.tex` (publication-ready table)

The dataset, the generator scripts, and the mechanism-selection protocols referenced in main §6 (search spaces, objective function, cycle-assignment selection) are available from the author on request pending public deposit. The spectral companion artifacts of [B] are archived separately on Zenodo (DOI: [10.5281/zenodo.19893371](https://doi.org/10.5281/zenodo.19893371)).

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